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BodyBalance AI

Health, activity & recovery support

Connect symptoms, health history,
and optional body-image analysis
for guidance and professional review.



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What is BodyBalance AI?

Health, activity, and recovery decision support

BodyBalance AI is a proposed platform that combines daily physical and mental check-ins, symptoms, health history, and optional posture-image analysis to support personalized activity and recovery guidance.



Images



Symptoms



Daily condition



History

The main concept

Symptoms

Record discomfort, pain location,
and how you feel today.

History

Connect previous symptoms, activities, and health reports.

Image analysis

Use optional body images for posture analysis and affected-area
documentation.

A connected summary for activity, recovery, and professional review

Planned decision support. Diagnosis and treatment remain with professionals.

Problem Definition

Prototype: the shoulder-pain demo

1 The situation

A student develops shoulder pain after activity.

2 The problem

Her photo, symptoms, and history are separate, leaving an unclear next step.

3 Why automation?

Connect these inputs and track changes over time to support guidance and professional review.

Live demo: bodybalance-ai-proto-uwso.bolt.host

Fictional demo data. Simulated observations.

Target Users & System Stakeholders



Active user

Track symptoms and recovery without switching between separate records.



Coach

Review authorized activity and recovery information to support coaching decisions.



Professional

Review an organized history and assessment summary with the user’s permission.

Users control which professionals can access their information.

Market study: strong players, one gap

Competitor	Strong at	Gap for BodyBalance AI to fill
MyFitnessPal	Nutrition and calorie tracking	No pain assessment or recovery decision support
Freeletics	Adaptive AI workouts	Plans from goals, not from today's pain or symptoms
Hinge Health	Musculoskeletal care and rehabilitation	Specialised clinical care, not daily fitness and recovery choices
Google Health / Fitbit	Wearable activity, sleep and stress tracking	Tracks metrics; no pain mapping or visual body analysis

Our comparison helped us focus on a connected assessment journey: bringing daily condition, symptoms, history, and optional images into one summary..

A focused first version



Technical feasibility

Start with check-ins, pain logs, history, and summaries. Test image analysis separately.



Operational feasibility

Use clear permissions and a simple workflow. Involve physiotherapists in review.



Financial feasibility

Assess development, hosting, AI processing, security, and maintenance costs.

Expansion depends on testing, suitable data, resources, and professional review.

Business Model Canvas

Key partners

Healthcare professionals, universities, gyms and sports clubs, cloud and AI providers

Key activities

Combine assessment inputs, provide guidance, track recovery, and prepare professional summaries.

Key resources

Assessment data, AI models, guidance and safety rules, secure hosting, and professional expertise.

Value propositions

Help users understand their current condition by connecting symptoms, health history, and optional posture analysis into personalized activity and recovery guidance, progress tracking, and summaries for professional review.

Relationships

Onboarding, reminders, feedback, user-controlled sharing

Channels

Mobile app, campus groups, social media, gym partners

Customer segments

Initial focus: Active students and fitness users.
Professional users: Coaches and healthcare professionals.
Potential future users: Older adults, people with disabilities, and caregivers

Cost structure

Development, cloud and AI, security, testing, consultation, marketing, support

Revenue streams

Free basic tier, premium subscription, coach and professional plans, organisational licences

How we identified requirements



Market comparison

Identify existing services and define a focused starting point.



Persona & use-case analysis

Turn each user need into a system action and expected outcome.



Team review

Refine wording and check consistency across the requirements and models.

The prototype helps us explain the proposed experience and gather future feedback.

Key functional requirements

01	**Daily check-in**	Record physical and mental condition before guidance.
02	**Pain assessment**	Select up to 3 pain zones and rate each from 1–10.
03	**Image analysis**	Accept JPEG / PNG posture images up to 10 MB.
04	**Safety check**	Check predefined red flags before a routine activity plan.
05	**History & access**	Store assessment records and control professional access.

Planned system requirements. The demo does not implement every function.

Quality requirements and verification



Performance target

Process a valid posture image within 5 seconds.
Verify with timed test runs.



Access control

Permit record access only to the owner or an authorized professional. Test denied access.



Safety boundary

Block routine exercise guidance
when a predefined red flag is detected. Test each rule.

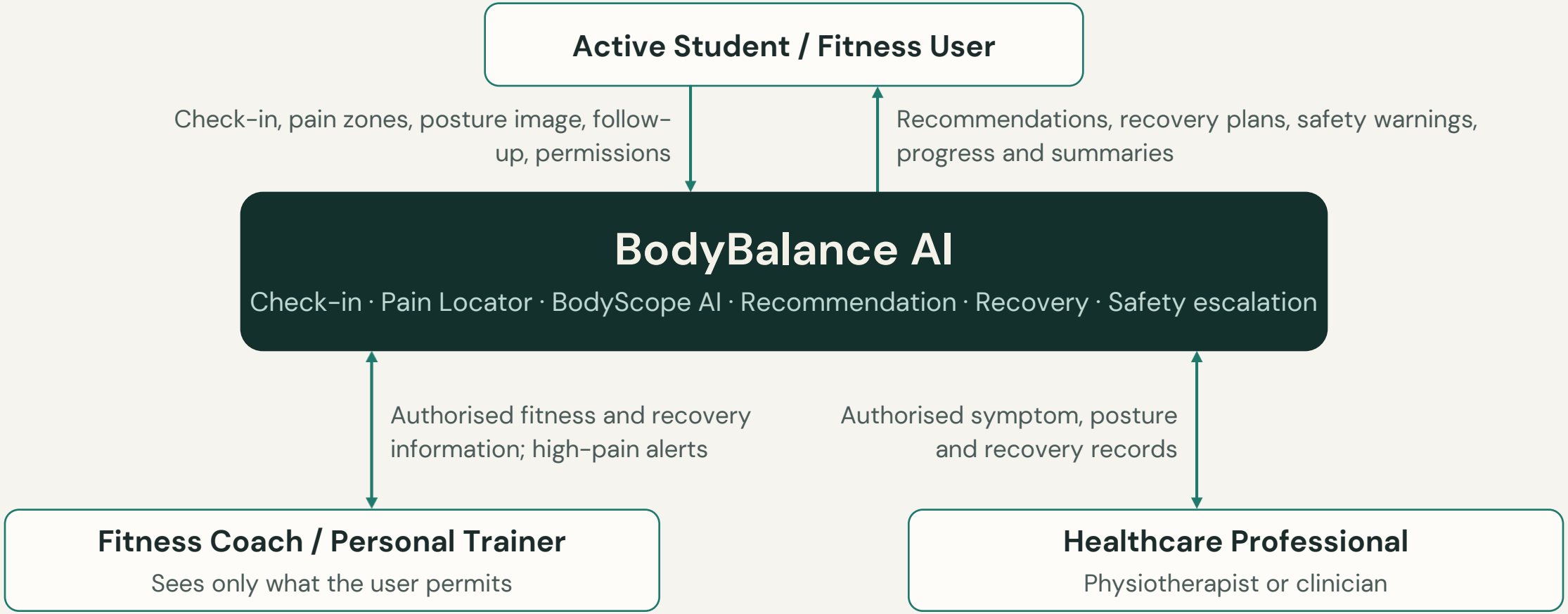
These are acceptance targets for the planned system, not measured demo results.

Use cases: five goal-level titles

ID	Use case	Includes	Priority
UC-01	Manage User Account	Register, log in, profile	Must
UC-02	Perform Body Assessment	Check-in, pain, posture	Must
UC-03	Generate Safe Activity Recommendation	Safety check, guidance, referral	Must
UC-04	Manage Recovery and Progress	Recovery plan, follow-up, history	Must
UC-05	Manage Professional Access	Permissions, summary, coach and clinician review	Should

UC-03 is the most complex: the safety check runs before any recommendation. Each has a detailed use case diagram, an activity diagram and a sequence diagram.

Context diagram: the system boundary



Saudi Vision 2030 alignment

Vision 2030 area	How BodyBalance AI aim to contribute
Quality of Life Program	Safer, more regular physical activity, starting with students and gym users
Health Sector Transformation	Digital, preventive guidance with clear referral to professional care
Digital economy and entrepreneurship	A student-founded AI product with freemium and subscription revenue
Data and AI direction	Consent, permission and privacy controls built in from the start
Good Health and Well-being	Support physical and mental well-being through symptom tracking, activity and recovery guidance, and organized information for professional review.

Applicable Saudi privacy and digital-health requirements are to be confirmed before launch.

Project management: shared work

Initial idea

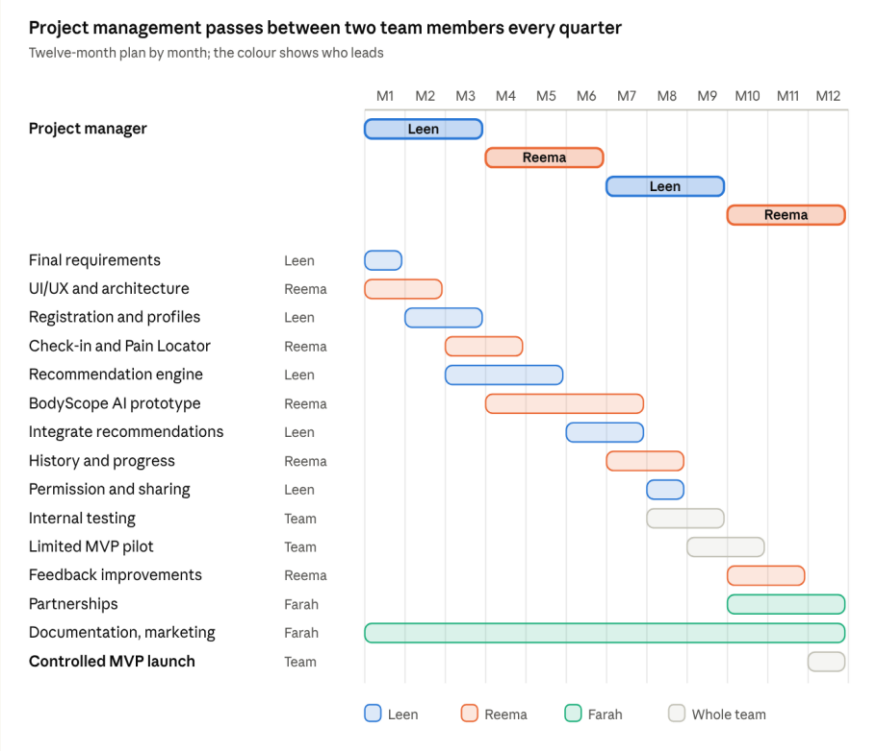
Leen and Reema developed the concept.
Farah joined afterward and contributed to its refinement.

Shared contributions

All three members contributed to the market study, personas, selected use cases, requirements, and UML models.

How we work

Divide tasks, review each other's work,
and combine contributions into consistent documents.



Next steps

Validate
Run interviews and a usability test; review the red-flag list with a healthcare adviser

Test
Technical testing of the recommendation engine and BodyScope AI against the SMART targets

Confirm
Privacy and digital-health requirements; update Lab 4 diagrams to the final requirement IDs

Thank you

Understand today. Support the next step.